

Blind Calibration of Timing Offsets for Four-Channel Time-Interleaved ADCs

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Abstract—In this paper, we describe a blind calibration method for timing mismatches in a four-channel time-interleaved analog-to-digital converter (ADC). The proposed method requires that the input signal should be slightly oversampled. This ensures that there exists a frequency band around the zero frequency where the Fourier transforms of the four ADC subchannels contain only three alias components, instead of four. Then the matrix power spectral density (PSD) of the ADC subchannels is rank deficient over this frequency band. Accordingly, when the timing offsets are known, we can construct a filter bank that nulls the vector signal at the ADC outputs. We employ a parametrization of this filter bank to develop an adaptive null steering algorithm for estimating the ADC timing offsets. The null steering filter bank employs seven fixed finite-impulse response filters and three unknown timing offset parameters which are estimated by using an adaptive stochastic gradient technique. A convergence analysis is presented for the blind calibration method. Numerical simulations for a bandlimited white noise input and for inputs containing several sinusoidal components demonstrate the effectiveness of the proposed technique.

Index Terms—Adaptive null steering filter bank, blind calibration, convergence of adaptive algorithms, Farrow structure, time-interleaved analog-to-digital converter (ADC).

I. INTRODUCTION

APPLICATIONS requiring analog-to-digital converters (ADCs) operating at very high rates have prompted the investigation in recent years [1]–[6] of time-interleaved ADC architectures of the type originally discussed in [7]. In such architectures, a fast ADC is obtained by combining slower ADCs operating in parallel. Ideally, the slow ADCs should have the same gain, and should sample the signal with timing phases that are uniformly spaced in time. Unfortunately, due to component mismatches and timing jitter effects, this requirement is impossible to achieve in practice. Note in this respect that while the timing issue can be solved by placing a common sample-and-hold (S/H) block in front of the slow ADCs, this S/H needs to operate at the overall rate of the fast ADC, which places unrealistic constraints on its settling time. So, usually each slow ADC has its own S/H block, which creates timing mismatches between the different ADCs. An intermediate choice which was adopted for the four-channel time-interleaved ADC described in [8] consists of dividing the

slow ADCs into groups, each with its own S/H circuit. The gain and timing mismatches arising in time-interleaved ADCs can degrade performance significantly [9]–[11], so they need to be estimated and corrected. The correction of gain and timing mismatches can be accomplished by tuning the operation of the parallel ADCs in the analog domain, but this is difficult to accomplish without introducing significant random jitter. Instead, it is preferable to perform the correction directly in the digital domain by viewing the samples produced by the time-interleaved ADC as periodic nonuniform samples of the signal and then using digital filter banks [12], [13] to synthesize the uniformly sampled signal. Note in this respect that while the technique described in [12] requires that the overall ADC should operate almost at twice the Nyquist rate of the input signal, the methods described in [13] and [14] requires only a small fraction, say around 10 to 20%, of excess samples.

Of course, in order to be able to correct gain and timing mismatches, one must first be able to estimate them. This can be accomplished either actively (in the foreground) by injecting a test signal in the ADC and then estimating mismatches from the resulting ADC output [15], [16], or blindly (in the background) by performing measurements while the ADC is in normal operation. The advantage of blind calibration methods is that they do not require special calibration phases, and also that they can track mismatch changes. Several blind calibration techniques for time-interleaved ADCs have been proposed recently [17]–[21]. Like blind equalization methods of digital communications [22], these techniques usually require knowledge of one key feature of the input signal which is exploited in the calibration process. Ideally, the required knowledge should be minimal, and the computational complexity of the calibration method should remain modest. From this perspective, the expectation-maximization technique proposed in [19] requires knowledge of the PSD of the input signal, which is probably unrealistic, whereas the blind method of [18] has a high computational complexity since it involves an iterative minimization requiring the evaluation of long FFTs at each iteration. The two-channel blind calibration methods described in [17] and [20] both require that the input signal should be oversampled and have a relatively low computational complexity, since they require only the implementation of two or three digital finite-impulse response (FIR) filters, respectively. However [17] requires a fraction of excess samples larger than 50%, whereas [20] only needs a small amount of oversampling.

We consider here the extension of the method presented in [20] to the case of four-channel time-interleaved ADCs. For simplicity, we focus on the estimation of timing offsets and assume that the gains are calibrated separately. The method of [20] relied on the observation that in the two-channel case,

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when the input signal is slightly oversampled, there exists a frequency band around the zero frequency which is alias free. By low-pass filtering the ADC output signals to this frequency band, this allows the formulation of the calibration problem as a conventional time delay estimation problem. For the case of four sampling channels, when the input signal is oversampled, there exists a frequency band around the zero frequency where the Fourier transforms of the ADC outputs contain only three alias components, instead of four normally. This implies that the matrix power-spectral density (PSD) of the ADC subchannels is rank deficient over this frequency band. It is therefore possible to construct a filter bank that nulls out the vector ADC output over this frequency band. A parametrization of the filter bank is obtained in terms of the timing offsets. Since the timing offsets arising in time-interleaved ADCs are typically small (of the order of 1 to 2% of the sampling period), we employ a first-order Farrow filter structure [23], ([24], Sec. 7.6) to implement the null steering filter bank. The resulting filter bank employs seven fixed FIR filters that low-pass filter the ADC outputs to the reduced alias region, apply $1/4$, $1/2$ and $3/4$ fractional delays, and differentiate the resulting signals. The three unknown timing offset parameters are estimated adaptively by a stochastic gradient technique. A convergence analysis of the adaptive blind calibration method is presented. The analysis relies on the stochastic averaging/ordinary differential equation (ODE) method of [25]. By identifying a Lyapunov function for the ODE used to analyze the adaptive algorithm, and performing a quadratic approximation of this function, we show that the estimates of the timing offsets converge approximately to their correct values. This analysis is confirmed by numerical simulations for a bandlimited white noise input and inputs containing several sinusoidal components.

The paper is organized as follows. Section II describes the calibration problem for a four-channel time-interleaved ADC and describes a Farrow digital filtering structure which can be used to correct mismatches after they are estimated. The blind calibration method and adaptive filtering structure we employ are presented in Section III. The convergence of the blind calibration technique is analyzed in Section IV, and Section V contains simulation results. Finally Section VI presents conclusions and suggestions for further research.

II. PROBLEM DESCRIPTION

Let $x_c(t)$ be a bandlimited signal with bandwidth B , so that its continuous-time Fourier transform

$$X_c(j\Omega) = \int_{-\infty}^{\infty} x_c(t) e^{-j\Omega t} dt \quad (2.1)$$

is zero for $|\Omega| > B$. Then, provided the sampling frequency $f_s = 1/T_s$ is greater than the Nyquist frequency $f_N = B/\pi$, $x_c(t)$ can be reconstructed exactly from its samples $x(n) = x_c(nT_s)$.

Instead of using a single ADC to sample the signal, we employ four time-interleaved ADCs with sampling period $T_i = 4T_s$ and relative sampling phase offset $T_s = T_i/4$ as shown in

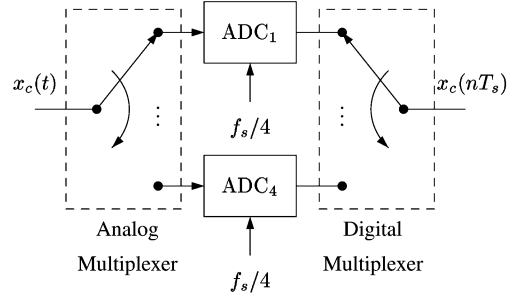


Fig. 1. Block diagram of a time-interleaved ADC.

Fig. 1. Unfortunately, due to timing offsets, quantization errors and thermal noise effects, the samples obtained at the outputs of the four sampling channels can be expressed as

$$z_i(m) = x_i(m) + v_i(m) \quad (2.2)$$

for $1 \leq i \leq 4$, where the noises $v_i(m)$ are independent zero-mean white Gaussian noises (WGNs) with variance $N_0/2$, and

$$\begin{aligned} x_1(m) &= x_c(mT_i) \\ x_2(m) &= x_c(mT_i + (1 + \delta_1)T_s) \\ x_3(m) &= x_c(mT_i + (2 + \delta_2)T_s) \\ x_4(m) &= x_c(mT_i + (3 + \delta_3)T_s) \end{aligned} \quad (2.3)$$

where for $1 \leq j \leq 3$, $\delta_j T_s$ represents the timing offset mismatch between the $j + 1$ th sampling channel and the first channel. Note that the time-interleaved sequences $x_i(m)$ with $1 \leq i \leq 4$ can be viewed as obtained from the fast sampled sequence $x(n)$ by first applying a noninteger time shift $(i - 1) + \delta_{i-1}$ with $\delta_0 = 0$ and then downsampling by a factor of 4, so that $x_i(m)$ can be expressed as

$$x_i(m) = (f_i * x)(4m) \quad (2.4)$$

with

$$f_i(n) = \frac{\sin(\pi(n + (i - 1) + \delta_{i-1}))}{\pi(n + (i - 1) + \delta_{i-1})} \quad (2.5)$$

where $*$ denotes the convolution operation.

For $1 \leq i \leq 4$, the discrete-time Fourier transform (DTFT) of $f_i(n)$ is

$$F_i(e^{j\omega}) = e^{j\omega(i-1+\delta_{i-1})} \quad (2.6)$$

for $-\pi < \omega < \pi$, so the four-channel time-interleaved ADC with timing offsets admits the digital filter bank model shown in Fig. 2.

Let

$$\mathbf{X}_I(e^{j\omega}) = \begin{bmatrix} X_1(e^{j\omega}) \\ X_2(e^{j\omega}) \\ X_3(e^{j\omega}) \\ X_4(e^{j\omega}) \end{bmatrix} \quad (2.7)$$

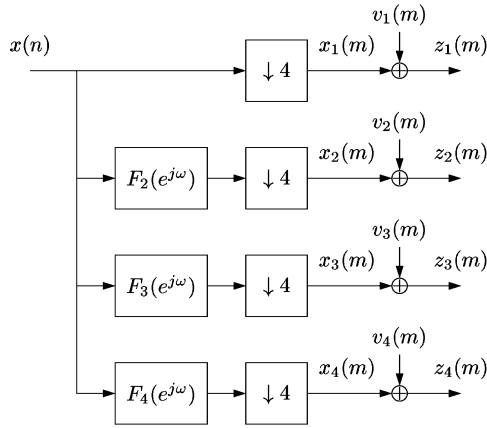


Fig. 2. Analysis filter bank model of the ADC.

$$\mathbf{X}_A(e^{j\omega}) = \begin{bmatrix} X(e^{j\frac{\omega}{4}}) \\ X(e^{j(\frac{\omega}{4} - \frac{\pi}{2})}) \\ X(e^{j(\frac{\omega}{4} - \pi)}) \\ X(e^{j(\frac{\omega}{4} - \frac{3\pi}{2})}) \end{bmatrix} \quad (2.8)$$

denote respectively the vectors formed by the DTFTs of the channel sampling sequences $x_i(m)$, and by the alias components of the DTFT of the fast sampled sequence $x(n)$. We have [26, ch. 5]

$$\mathbf{X}_I(e^{j\omega}) = \frac{1}{4} \mathbf{M}(e^{j\omega}) \mathbf{X}_A(e^{j\omega}) \quad (2.9)$$

where $\mathbf{M}(e^{j\omega})$, shown in (2.10) at the bottom of the page, denotes the alias component matrix of the analysis filter bank. Let

$$\boldsymbol{\delta} = [\delta_1 \quad \delta_2 \quad \delta_3]^T \quad (2.11)$$

denote the vector formed by the relative timing mismatches of channels 2, 3, and 4 with respect to the first channel. Taking into account expression (2.6) for the filters $F_i(e^{j\omega})$, and making the $\boldsymbol{\delta}$ dependence of $\mathbf{M}$ explicit, we find that $\mathbf{M}(e^{j\omega}, \boldsymbol{\delta})$ admits the factorization

$$\mathbf{M}(e^{j\omega}, \boldsymbol{\delta}) = \mathbf{D}(e^{j\omega}, \boldsymbol{\delta}) \mathbf{V}(\boldsymbol{\delta}) \quad (2.12)$$

where

$$\mathbf{D}(e^{j\omega}, \boldsymbol{\delta}) \triangleq \text{diag} \left\{ 1, e^{j\frac{\omega}{4}(1+\delta_1)}, e^{j\frac{\omega}{4}(2+\delta_2)}, e^{j\frac{\omega}{4}(3+\delta_3)} \right\} \quad (2.13)$$

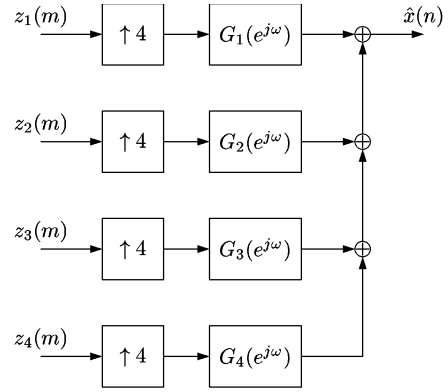


Fig. 3. Filter bank synthesis of the ADC output.

is a diagonal matrix and

$$\mathbf{V}(\boldsymbol{\delta}) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & u_1 & u_1^2 & u_1^3 \\ 1 & u_2 & u_2^2 & u_2^3 \\ 1 & u_3 & u_3^2 & u_3^3 \end{bmatrix} \quad (2.14)$$

is a Vandermonde matrix depending only on $\boldsymbol{\delta}$, with

$$u_i = e^{-j\frac{\pi}{2}(i+\delta_i)} \quad (2.15)$$

for $1 \leq i \leq 3$. By inverting the matrix $\mathbf{M}(e^{j\omega})$ in (2.9), or equivalently, by expressing the analysis filters $F_i(e^{j\omega})$ in terms of their polyphase components, and inverting the polyphase matrix, we find [26, ch. 5] that there exists synthesis filters $G_i(e^{j\omega})$ such that

$$X(e^{j\omega}) = \sum_{i=1}^4 G_i(e^{j\omega}) X_i(e^{4j\omega}) \quad (2.16)$$

so $x(n)$ can be reconstructed exactly from the sampled sequences $x_i(m)$ by first upsampling them by a factor of 4 and then passing them through the synthesis filter bank $\{G_i(e^{j\omega}), 1 \leq i \leq 4\}$ as shown in Fig. 3. When noise is present, so the inputs of the filter bank are the noisy observations $z_i(m)$, the output signal $\hat{x}(n)$ is a noise-corrupted form of the signal $x(n)$.

Unfortunately, when the delays δ_i , $1 \leq i \leq 3$ are arbitrary, even if one attempts to exploit the structure (2.12)–(2.14) of $\mathbf{M}(e^{j\omega}, \boldsymbol{\delta})$, it has proven impossible thus far to derive closed-form expressions for the filters $G_i(e^{j\omega})$. These filters must therefore be obtained numerically, and a procedure to do

$$\mathbf{M}(e^{j\omega}) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ F_2(e^{j\frac{\omega}{4}}) & F_2(e^{j(\frac{\omega}{4} - \frac{\pi}{2})}) & F_2(e^{j(\frac{\omega}{4} - \pi)}) & F_2(e^{j(\frac{\omega}{4} - \frac{3\pi}{2})}) \\ F_3(e^{j\frac{\omega}{4}}) & F_3(e^{j(\frac{\omega}{4} - \frac{\pi}{2})}) & F_3(e^{j(\frac{\omega}{4} - \pi)}) & F_3(e^{j(\frac{\omega}{4} - \frac{3\pi}{2})}) \\ F_4(e^{j\frac{\omega}{4}}) & F_4(e^{j(\frac{\omega}{4} - \frac{\pi}{2})}) & F_4(e^{j(\frac{\omega}{4} - \pi)}) & F_4(e^{j(\frac{\omega}{4} - \frac{3\pi}{2})}) \end{bmatrix} \quad (2.10)$$

so by constructing a least-squares FIR inverse of the polyphase matrix of the analysis filter bank is described in [13] (see also [12]–[14]). However, in time-interleaved ADCs, the timing offsets are usually rather small, say of the order of 1 to 2% of the sampling period [4], [27], and in this case it is shown in Appendix A that the synthesis filters admit the first-order Farrow approximation

$$\begin{aligned}
 G_1(e^{j\omega}, \delta) &\approx 1 - \frac{\pi}{4} \{ (\delta_1 - \delta_3) S_e(e^{j\omega}) \\
 &\quad + j(\delta_1 + \delta_3) S_o(e^{j\omega}) + j\delta_2 S_e(e^{j\omega}) S_o(e^{j\omega}) \} \\
 G_2(e^{j\omega}, \delta) &\approx \left[1 - j\delta_1 \Psi(e^{j\omega}) - \frac{\pi}{4} \{ \delta_2 S_e(e^{j\omega}) \right. \\
 &\quad \left. + j\delta_2 S_o(e^{j\omega}) + j\delta_3 S_e(e^{j\omega}) S_o(e^{j\omega}) \} \right] e^{-j\omega} \\
 G_3(e^{j\omega}, \delta) &\approx \left[1 - j\delta_2 \Psi(e^{j\omega}) - \frac{\pi}{4} \{ -(\delta_1 - \delta_3) S_e(e^{j\omega}) \right. \\
 &\quad \left. + j(\delta_1 + \delta_3) S_o(e^{j\omega}) \} \right] e^{-2j\omega} \\
 G_4(e^{j\omega}, \delta) &\approx \left[1 - j\delta_3 \Psi(e^{j\omega}) - \frac{\pi}{4} \{ -\delta_2 S_e(e^{j\omega}) \right. \\
 &\quad \left. + j\delta_2 S_o(e^{j\omega}) + j\delta_1 S_e(e^{j\omega}) S_o(e^{j\omega}) \} \right] e^{-3j\omega}
 \end{aligned} \quad (2.17)$$

where the functions $S_e(e^{j\omega})$, $S_o(e^{j\omega})$, and $\Psi(e^{j\omega})$ are given by

$$S_e(e^{j\omega}) = \begin{cases} -1, & 0 \leq |\omega| < \pi/2 \\ 1, & \pi/2 \leq |\omega| < \pi \end{cases} \quad (2.18)$$

$$S_o(e^{j\omega}) = \text{sgn}(\omega) \quad (2.19)$$

for $-\pi \leq \omega < \pi$, and

$$\Psi(e^{j\omega}) = \begin{cases} \omega - \frac{\pi}{4} \text{sgn}(\omega), & 0 \leq |\omega| < \pi/2 \\ \omega - \frac{3\pi}{4} \text{sgn}(\omega), & \pi/2 \leq |\omega| < \pi. \end{cases} \quad (2.20)$$

These functions are sketched in Fig. 4.

Note that since $S_e(\omega)$ is even, while $S_o(e^{j\omega})$ and $\Psi(e^{j\omega})$ are odd, the filter G_i satisfies

$$G_i(e^{-j\omega}, \delta) = (G_i(e^{j\omega}, \delta))^* \quad (2.21)$$

so its impulse response is real. Instead of implementing directly the filters $G_i(e^{j\omega}, \delta)$, it is convenient to implement the fixed filters $S_e(e^{j\omega})$, $jS_o(e^{j\omega})$ and $j\Psi(e^{j\omega})$, and then apply a linear weighting to their outputs (with the exact δ_i s replaced by their estimates $\hat{\delta}_i$) to form the filters G_i according to (2.17). However, the filters $S_e(e^{j\omega})$, $jS_o(e^{j\omega})$ and $j\Psi(e^{j\omega})$ have phase discontinuities at $0, \pm\pi/2$ and $\pm\pi$, so long FIR filters are required to implement them. In the two-channel case, the correction filters already contained discontinuities at 0 and $\pm\pi$ [17], [20], but these discontinuities were of little consequence since typical input signals do not have a dc component and are oversampled, so they do not have any frequency content at $\pm\pi$. For the four-channel case, the correction filter discontinuity at $\pm\pi/2$ is more problematic, since signals usually have some frequency content at these frequencies, and thus long FIR filters are required to obtain a precise ADC correction.

To estimate the unknown delays δ_i , $1 \leq i \leq 3$, we employ a blind calibration approach, where the δ_i 's are estimated while the ADC is operating normally on an arbitrary unknown signal.

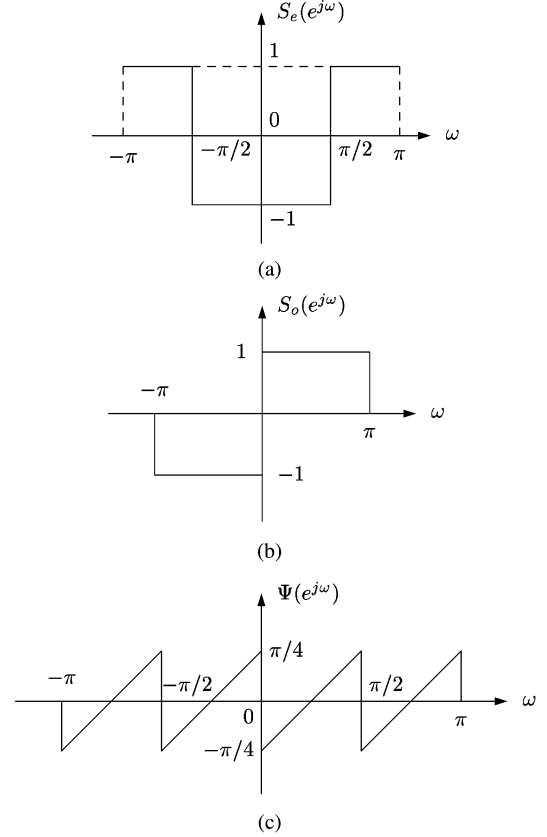


Fig. 4. Plots of the functions (a) $S_e(e^{j\omega})$, (b) $S_o(e^{j\omega})$, and (c) $\Psi(e^{j\omega})$.

III. BLIND CALIBRATION METHOD

A. Reduced Aliasing Property

Like the blind calibration approach employed in [20] for the two-channel case, we assume that the continuous-time signal $x_c(t)$ is slightly oversampled by the ADC, so that the DTFT $X(e^{j\omega})$ of the sampled sequence $x(n) = x_c(nT_s)$ is zero in a band of width $\alpha\pi$ centered about $\pm\pi$, i.e.,

$$X(e^{j\omega}) = 0, \quad \text{for } |\omega \pm \pi| < \alpha\pi \quad (3.1)$$

where

$$\alpha \triangleq (f_s - f_N)/f_s \quad (3.2)$$

denotes the proportion of excess samples. We assume temporarily that $\alpha < 1/4$, which represents the more difficult case where none of the four time-interleaved ADCs is redundant. In this case, as shown in Fig. 5, in the frequency band $[-4\alpha\pi, 4\alpha\pi]$ the DTFTs $X_i(e^{j\omega})$ of the sampled sequences contain only three alias components of the signal $x(n)$, so that (2.9), (2.12) for the vector $\mathbf{X}_I(e^{j\omega})$ becomes

$$\mathbf{X}_I(e^{j\omega}) = \frac{1}{4} \mathbf{M}_R(e^{j\omega}, \delta) \begin{bmatrix} X(e^{j(\frac{\omega}{4} + \frac{\pi}{2})}) \\ X(e^{j\frac{\omega}{4}}) \\ X(e^{j(\frac{\omega}{4} - \frac{\pi}{2})}) \end{bmatrix} \quad (3.3)$$

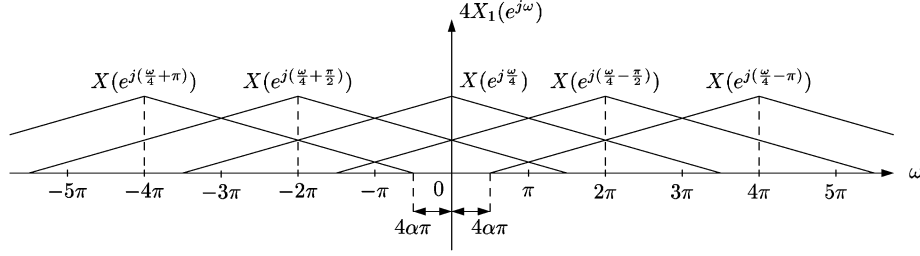


Fig. 5. Missing alias component property of oversampled signals.

for $0 < |\omega| < 4\alpha\pi$, where $\mathbf{M}_R(e^{j\omega}, \delta)$ is the 4×3 reduced alias matrix

$$\begin{aligned} \mathbf{M}_R(e^{j\omega}, \delta) &= \begin{bmatrix} 1 & 1 & 1 \\ F_2(e^{j(\frac{\omega}{4} + \frac{\pi}{2})}) & F_2(e^{j\frac{\omega}{4}}) & F_2(e^{j(\frac{\omega}{4} - \frac{\pi}{2})}) \\ F_3(e^{j(\frac{\omega}{4} + \frac{\pi}{2})}) & F_3(e^{j\frac{\omega}{4}}) & F_3(e^{j(\frac{\omega}{4} - \frac{\pi}{2})}) \\ F_4(e^{j(\frac{\omega}{4} + \frac{\pi}{2})}) & F_4(e^{j\frac{\omega}{4}}) & F_4(e^{j(\frac{\omega}{4} - \frac{\pi}{2})}) \end{bmatrix} \\ &= \mathbf{D}(e^{j\omega}, \delta) \mathbf{V}_R(\delta) \end{aligned} \quad (3.4)$$

with

$$\mathbf{V}_R(\delta) = \begin{bmatrix} 1 & 1 & 1 \\ u_1^{-1} & 1 & u_1 \\ u_2^{-1} & 1 & u_2 \\ u_3^{-1} & 1 & u_3 \end{bmatrix} \quad (3.5)$$

for $|\omega| < 4\alpha\pi$, where u_i is given by (2.15) for $1 \leq i \leq 3$.

B. Null-Steering Filter Bank

Since the matrix $\mathbf{V}_R(\delta)$ has size 4×3 , we can always find a vector

$$\mathbf{c}^T(\delta) = [c_1(\delta) \quad c_2(\delta) \quad c_3(\delta) \quad c_4(\delta)] \quad (3.6)$$

in its left-null space, so that

$$\mathbf{c}^T(\delta) \mathbf{V}_R(\delta) = 0. \quad (3.7)$$

In the following, $\mathbf{c}^T(\delta)$ will be called the null-steering vector. Note that the vector $\mathbf{c}^T(\delta)$ is determined only up to an arbitrary scaling parameter, which is fixed by requiring

$$|\mathbf{c}(\delta)|^2 \triangleq \sum_{i=1}^4 c_i^2(\delta) = 4. \quad (3.8)$$

The choice of normalization constant is arbitrary and does not affect our analysis, as long as it is independent of the delay vector δ . Let

$$\hat{\delta} \triangleq [\hat{\delta}_1 \quad \hat{\delta}_2 \quad \hat{\delta}_3]^T \quad (3.9)$$

denote a vector of estimates of the actual delays δ_j with $1 \leq j \leq 3$, and consider the low-pass filter

$$H_{\text{LP}}(e^{j\omega}) = \begin{cases} 1, & 0 < |\omega| < 4\alpha\pi \\ 0, & 4\alpha\pi < |\omega| < \pi. \end{cases} \quad (3.10)$$

The null-steering filter that we use to estimate the true delay vector δ has the structure

$$\mathbf{H}(e^{j\omega}, \hat{\delta}) = \mathbf{c}^T(\hat{\delta}) \mathbf{K}(e^{j\omega}, \hat{\delta}) \quad (3.11)$$

where the filter bank

$$\mathbf{K}(e^{j\omega}, \hat{\delta}) = \text{diag}\{K_i(e^{j\omega}, \hat{\delta}_{i-1}), 1 \leq i \leq 4\} \quad (3.12)$$

is formed by the low-pass fractional delay filters

$$K_i(e^{j\omega}, \hat{\delta}_{i-1}) = e^{-j\omega(i-1+\hat{\delta}_{i-1})/4} H_{\text{LP}}(e^{j\omega}) \quad (3.13)$$

with $\hat{\delta}_0 = 0$.

To explain the structure of the null-steering filter bank $\mathbf{H}(e^{j\omega}, \hat{\delta})$, let $\mathbf{h}(m, \hat{\delta})$ denote its impulse response, and write as

$$\mathbf{z}(m) = [z_1(m) \quad z_2(m) \quad z_3(m) \quad z_4(m)]^T \quad (3.14)$$

the vector signal formed by the ADC outputs. By passing $\mathbf{z}(m)$ through $\mathbf{h}(m, \hat{\delta})$, we obtain the error signal

$$e(m, \hat{\delta}) = \mathbf{h}(m, \hat{\delta}) * \mathbf{z}(m). \quad (3.15)$$

As indicated in (2.2), the ADC output vector $\mathbf{z}(m)$ can be divided between signal and noise components as

$$\mathbf{z}(m) = \mathbf{x}_I(m) + \mathbf{v}_I(m) \quad (3.16)$$

with

$$\begin{aligned} \mathbf{x}_I(m) &\triangleq [x_1(m) \quad x_2(m) \quad x_3(m) \quad x_4(m)]^T \\ \mathbf{v}_I(m) &\triangleq [v_1(m) \quad v_2(m) \quad v_3(m) \quad v_4(m)]^T. \end{aligned} \quad (3.17)$$

Accordingly, the error $e(m, \hat{\delta})$ can be decomposed into signal and noise components as

$$e(m, \hat{\delta}) = r(m, \hat{\delta}) + w(m, \hat{\delta}) \quad (3.18)$$

where

$$\begin{aligned} r(m, \hat{\delta}) &\triangleq \mathbf{h}(m, \hat{\delta}) * \mathbf{x}_I(m) \\ w(m, \hat{\delta}) &\triangleq \mathbf{h}(m, \hat{\delta}) * \mathbf{v}_I(m). \end{aligned} \quad (3.19)$$

However, we have already observed that over the frequency band $[-4\alpha\pi, 4\alpha\pi]$, the Fourier transform $\mathbf{X}_I(e^{j\omega})$ of the signal vector $\mathbf{x}_I(m)$ can be expressed in terms of the alias components of the fast sampled sequence as (3.3). Accordingly, by combining (3.3) with the structure (3.11)–(3.13) of the null steering filter bank, we find that Fourier transform of the signal component $r(m, \hat{\delta})$ of the error takes the form

$$R(e^{j\omega}, \hat{\delta}) = \frac{1}{4} \mathbf{c}^T(\hat{\delta}) \mathbf{E}(e^{j\omega}, \delta - \hat{\delta}) \mathbf{V}_R(\delta) \begin{bmatrix} X(e^{j(\frac{\omega}{4} + \frac{\pi}{2})}) \\ X(e^{j(\frac{\omega}{4})}) \\ X(e^{j(\frac{\omega}{4} - \frac{\pi}{2})}) \end{bmatrix} \quad (3.20)$$

with

$$\mathbf{E}(e^{j\omega}, \delta - \hat{\delta}) \triangleq \text{diag}\left\{e^{j\frac{\omega}{4}(\delta_{i-1} - \hat{\delta}_{i-1})}, 1 \leq i \leq 4\right\} \quad (3.21)$$

for $0 \leq |\omega| \leq 4\alpha\pi$, and

$$R(e^{j\omega}, \hat{\delta}) = 0 \quad (3.22)$$

for $4\alpha\pi \leq |\omega| \leq \pi$. The expression (3.20) has the interesting feature that when the estimated delay vector $\hat{\delta} = \delta$, since $\mathbf{E}(e^{j\omega}, \mathbf{0}) = \mathbf{I}_4$ and the vector $\mathbf{c}^T(\delta)$ in the left null-space of $\mathbf{V}_R(\delta)$, we have

$$R(e^{j\omega}, \delta) = 0 \quad (3.23)$$

for all ω . So when the estimated vector $\hat{\delta}$ parametrizing the null-steering filter bank coincides with the true delay vector δ , the signal component $r(m, \delta)$ of the error signal $e(m, \delta)$ is nulled out. Note that this feature holds for any input signal $x(n)$, as long as it is slightly oversampled and admits a Fourier transform.

This property suggests that by adjusting adaptively the estimated delay vector $\hat{\delta}$ so that the power of the error signal $e(m, \hat{\delta})$ is minimized, we will be able to estimate the unknown delay vector δ . However, for this approach to work, we must also ensure that the power of the noise component $w(m, \hat{\delta})$ is unaffected by the choice of $\hat{\delta}$. Since the noise vector $\mathbf{v}_I(m)$ is a zero-mean WGN with variance matrix $N_0/2\mathbf{I}_4$, the noise $w(m, \hat{\delta})$ is zero-mean wide-sense stationary (WSS) with PSD

$$\begin{aligned} S_w(e^{j\omega}, \hat{\delta}) &= \mathbf{H}(e^{j\omega}, \hat{\delta}) \mathbf{H}^H(e^{j\omega}, \hat{\delta}) \frac{N_0}{2} \\ &= \begin{cases} |\mathbf{c}(\hat{\delta})|^2 \frac{N_0}{2}, & 0 \leq |\omega| \leq 4\alpha\pi \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (3.24)$$

So $w(m, \hat{\delta})$ is a bandlimited zero-mean WGN, and when the normalization (3.8) is satisfied, its PSD $S_w(e^{j\omega}) = 2N_0$ over the passband $[-4\alpha\pi, 4\alpha\pi]$ is independent of the estimated delay $\hat{\delta}$. So from this perspective, the purpose of the normalization constraint (3.8) for the steering vector $\mathbf{c}^T(\hat{\delta})$ is to ensure that

the noise power at the output of the null-steering filter bank is insensitive to the choice of steering direction.

So in conclusion we have demonstrated in this section that the reduced aliasing property of the ADC outputs over the frequency band $[-4\alpha\pi, 4\alpha\pi]$ can be exploited by designing a null-steering filter bank that applies the null-steering vector $\mathbf{c}(\hat{\delta})$ to the outputs of a bank of low-pass fractional delay filters $K_i(e^{j\omega}, \hat{\delta}_{i-1})$ with $1 \leq i \leq 4$. This null-steering filter has the feature that the output signal component is zero when the delay vector is estimated correctly, and the output noise statistics are independent of the steering direction. This type of null steering structure is often employed in array processing [28], [29] when a vector signal impinging on a receiver array has a rank deficient covariance matrix,

Remark: Up to this point we have assumed that the oversampling factor $\alpha < 1/4$. When α exceeds $1/4$, the reduced aliasing property (3.3) holds over the entire frequency band $[-\pi, \pi]$, and in this case it is not necessary to include the low-pass-filter $H_{LP}(e^{j\omega})$ in the specification (3.13) of the delay filters $K_i(e^{j\omega}, \hat{\delta}_{i-1})$. Also, expressions (3.20) and (3.24) for the Fourier transform of $r(m, \hat{\delta})$ and the PSD of $w(m, \hat{\delta})$ hold over the entire frequency band $[-\pi, \pi]$, so $w(m, \hat{\delta})$ is a zero-mean WGN with variance $2N_0$.

C. Null-Steering Vector Parametrization

To implement the null-steering filter bank $\mathbf{H}(e^{j\omega}, \hat{\delta})$, we need first to obtain a closed-form expression for the vector $\mathbf{c}^T(\delta)$ satisfying matrix equation (3.6) with the normalization (3.8). To solve (3.6) we observe that since the complex numbers u_i given by (2.15) are located on the unit circle, the first and third columns of the matrix $\mathbf{V}_R(\delta)$ are complex conjugate of each other. Equation (3.6) can therefore be replaced by the real equation

$$\mathbf{c}^T(\delta) \mathbf{T}_R(\delta) = 0 \quad (3.25)$$

with

$$\mathbf{T}_R(\delta) \triangleq \begin{bmatrix} 1 & 1 & 0 \\ 1 & u_{1R} & u_{1I} \\ 1 & u_{2R} & u_{2I} \\ 1 & u_{3R} & u_{3I} \end{bmatrix} \quad (3.26)$$

where

$$\begin{aligned} u_{iR} &= \Re\{u_i\} = \cos\left(\frac{\pi}{2}(i + \delta_i)\right) \\ u_{iI} &= \Im\{u_i\} = -\sin\left(\frac{\pi}{2}(i + \delta_i)\right). \end{aligned} \quad (3.27)$$

To solve (3.6), we set $c_1(\delta) = 1$ and rewrite (3.6) as

$$-[1 \quad 1 \quad 0] = \mathbf{c}_R^T(\delta) \mathbf{U}_R(\delta) \quad (3.28)$$

with

$$\mathbf{c}_R^T(\delta) \triangleq [c_2(\delta) \quad c_3(\delta) \quad c_4(\delta)] \quad (3.29)$$

and

$$\mathbf{U}_R(\boldsymbol{\delta}) = \begin{bmatrix} 1 & u_{1R} & u_{1I} \\ 1 & u_{2R} & u_{2I} \\ 1 & u_{3R} & u_{3I} \end{bmatrix} = \begin{bmatrix} 1 & -\sin(\frac{\pi}{2}\delta_1) & -\cos(\frac{\pi}{2}\delta_1) \\ 1 & -\cos(\frac{\pi}{2}\delta_2) & \sin(\frac{\pi}{2}\delta_2) \\ 1 & \sin(\frac{\pi}{2}\delta_3) & \cos(\frac{\pi}{2}\delta_3) \end{bmatrix}. \quad (3.30)$$

By applying Cramer's rule, we find

$$\begin{aligned} c_2(\boldsymbol{\delta}) &= \left[-\sin\left(\frac{\pi}{2}\delta_2\right) + \cos\left(\frac{\pi}{2}\delta_3\right) + \cos\left(\frac{\pi}{2}(\delta_3 - \delta_2)\right) \right] / |\mathbf{U}_R| \\ c_3(\boldsymbol{\delta}) &= \left[-\cos\left(\frac{\pi}{2}\delta_1\right) - \cos\left(\frac{\pi}{2}\delta_3\right) + \sin\left(\frac{\pi}{2}(\delta_3 - \delta_1)\right) \right] / |\mathbf{U}_R| \\ c_4(\boldsymbol{\delta}) &= \left[\sin\left(\frac{\pi}{2}\delta_2\right) + \cos\left(\frac{\pi}{2}\delta_1\right) + \cos\left(\frac{\pi}{2}(\delta_1 - \delta_2)\right) \right] / |\mathbf{U}_R| \end{aligned} \quad (3.31)$$

and

$$|\mathbf{U}_R| = - \left[\cos\left(\frac{\pi}{2}(\delta_3 - \delta_2)\right) + \cos\left(\frac{\pi}{2}(\delta_2 - \delta_1)\right) + \sin\left(\frac{\pi}{2}(\delta_3 - \delta_1)\right) \right] \quad (3.32)$$

where $|\mathbf{U}_R|$ denotes the determinant of $\mathbf{U}_R$. Note that since the timing offsets δ_i are small, we can perform a first-order Taylor series approximation of $c_i(\boldsymbol{\delta})$ for $2 \leq i \leq 4$, which yields

$$\begin{aligned} c_2(\boldsymbol{\delta}) &\approx -1 + \frac{\pi}{4}(\delta_2 + \delta_3 - \delta_1) \\ c_3(\boldsymbol{\delta}) &\approx 1 - \frac{\pi}{2}(\delta_3 - \delta_1) \\ c_4(\boldsymbol{\delta}) &\approx -1 + \frac{\pi}{4}(-\delta_2 + \delta_3 - \delta_1) \end{aligned} \quad (3.33)$$

where in (3.33) we neglect quadratic terms of the form $\delta_i\delta_j$ for all i and j . Since all entries of the vector $\mathbf{c}(\boldsymbol{\delta})$ sum to zero, it is easy to check that

$$|\mathbf{c}(\boldsymbol{\delta})|^2 = 4 + O(\delta^2) \quad (3.34)$$

so the normalization (3.8) is satisfied, as long as we neglect quadratic terms of the entries of $\boldsymbol{\delta}$, which are represented here by $O(\delta^2)$. When the timing mismatches exceed the small values assumed here, it may be necessary to include higher order terms in the Taylor series expansion (3.23) of the coefficients $c_i(\boldsymbol{\delta})$ with $2 \leq i \leq 4$ and to perform a complete renormalization of the vector $\mathbf{c}(\boldsymbol{\delta})$ to ensure that the constraint (3.8) holds to the required level of precision.

D. Adaptive Filtering Structure

Instead of implementing directly the low-pass time delay filters $K_i(e^{j\omega}, \hat{\delta}_{i-1})$, which would require adjusting all filter taps simultaneously as the timing offsets δ_j are estimated, we use the fact that the timing offsets and their estimates $\hat{\delta}_j$ are small for $1 \leq j \leq 3$. Accordingly the filters admit the first-order Farrow approximation

$$K_i(e^{j\omega}, \hat{\delta}_{i-1}) \approx K_{i0}(e^{j\omega}) + \hat{\delta}_{i-1}K_{i1}(e^{j\omega}) \quad (3.35)$$

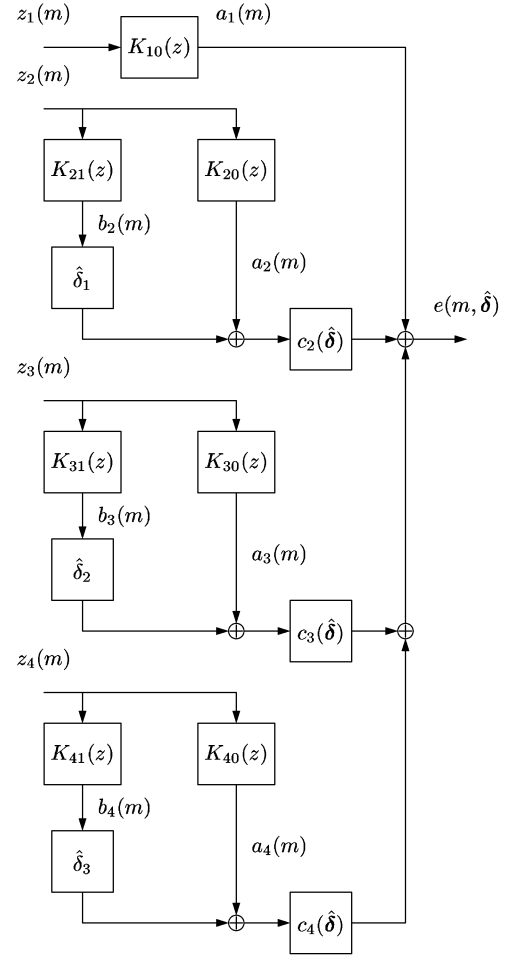


Fig. 6. Adaptive filtering structure.

where for $1 \leq i \leq 4$

$$K_{i0}(e^{j\omega}) = e^{-j\omega(i-1)/4} H_{LP}(e^{j\omega}) \quad (3.36)$$

applies a low-pass fractional delay equal to $(i-1)/4$, and

$$K_{i1}(e^{j\omega}) = \frac{-j\omega}{4} e^{-j\omega(i-1)/4} H_{LP}(e^{j\omega}) \quad (3.37)$$

combines a differentiator with a low-pass fractional delay equal to $(i-1)/4$. Note that if timing offsets are larger than the few percents of the sampling period we have assumed here, higher order terms would need to be included in the Farrow expansion (3.35).

We implement the adaptive null steering structure shown in Fig. 6. Let $k_{i0}(m)$ and $k_{i1}(m)$ denote the impulse responses of the filters $K_{i0}(e^{j\omega})$ and $K_{i1}(e^{j\omega})$. Then write as

$$\begin{aligned} a_i(m) &\triangleq k_{i0}(m) * z_i(m) \\ b_i(m) &\triangleq k_{i1}(m) * z_i(m) \end{aligned} \quad (3.38)$$

the signals obtained by passing the outputs $z_i(m)$ of the time-interleaved ADCs through the filters $K_{i0}(e^{j\omega})$ and $K_{i1}(e^{j\omega})$ as shown in Fig. 6. By using the Farrow approximation (3.35), the

error signal at the output of the null-steering filter bank can be expressed as

$$e(m, \hat{\delta}) = \sum_{i=1}^4 c_i(\hat{\delta})(a_i(m) + b_i(m)\hat{\delta}_{i-1}). \quad (3.39)$$

Note that $\hat{\delta}_0 = 0$, so the filter $K_{11}(e^{j\omega})$ does need not to be implemented.

We use the error signal $e(m, \hat{\delta})$ to adjust adaptively the estimated vector $\hat{\delta}$. Consider the objective function

$$J(\hat{\delta}) = \frac{1}{2} E[e^2(m, \hat{\delta})]. \quad (3.40)$$

If

$$\nabla_{\hat{\delta}} \triangleq [\partial/\partial\hat{\delta}_1 \quad \partial/\partial\hat{\delta}_2 \quad \partial/\partial\hat{\delta}_3]^T \quad (3.41)$$

the gradient of $J(\hat{\delta})$ with respect to $\hat{\delta}$ is given by

$$\nabla_{\hat{\delta}} J(\hat{\delta}) = E[e(m, \hat{\delta}) \nabla_{\hat{\delta}} e(m, \hat{\delta})] \quad (3.42)$$

with

$$\begin{aligned} \nabla_{\hat{\delta}} e(m, \hat{\delta}) &= \begin{bmatrix} c_2(\hat{\delta})b_2(m) \\ c_3(\hat{\delta})b_3(m) \\ c_4(\hat{\delta})b_4(m) \end{bmatrix} \\ &+ \sum_{i=2}^4 (a_i(m) + \hat{\delta}_{i-1}b_i(m)) \nabla_{\hat{\delta}} c_i(\hat{\delta}) \\ &\approx \begin{bmatrix} -b_2(m) \\ b_3(m) \\ -b_4(m) \end{bmatrix} \\ &+ \frac{\pi}{4} \begin{bmatrix} -a_2(m) - a_4(m) + 2a_3(m) \\ a_2(m) - a_4(m) \\ a_2(m) + a_4(m) - 2a_3(m) \end{bmatrix} \end{aligned} \quad (3.43)$$

where to derive the approximation (3.43) we employ the first-order expansion (3.33) of the coefficients $c_i(\hat{\delta})$ and neglect all quadratic or higher order products of the estimated delays. Applying a gradient descent minimization technique to $J(\hat{\delta})$ yields the iteration

$$\hat{\delta}(m+1) = \hat{\delta}(m) - \mu \nabla_{\hat{\delta}} J(\hat{\delta}(m)) \quad (3.44)$$

where μ denotes the iteration step size. Unfortunately, the expected value in the gradient expression (3.44) is not known, so we consider the stochastic gradient adaptive iteration [30]

$$\hat{\delta}(m+1) = \hat{\delta}(m) - \mu e(m, \hat{\delta}(m)) \nabla_{\hat{\delta}} e(m, \hat{\delta}(m)) \quad (3.45)$$

which is obtained by replacing the expected value by its instantaneous value in the gradient evaluation (3.44). Since the delays are unknown but small, the algorithm is initialized with

$$\hat{\delta}(0) = \mathbf{0}. \quad (3.46)$$

E. Error Signal Power

The expression (3.20) for the Fourier transform of the signal component $r(m, \hat{\delta})$ of the error signal provides a justification for the proposed adaptive null-steering scheme, since it depends continuously on $\hat{\delta}$ and $r(m, \hat{\delta})$ is nulled out when $\hat{\delta} = \delta$. As indicated earlier, this expression holds for any bandlimited signal with oversampling factor α .

However, to characterize precisely the shape of the cost function $J(\hat{\delta})$ in the vicinity of δ , it is necessary to specify the statistical behavior of the input signal $x_c(t)$. We assume that in addition to being bandlimited, $x_c(t)$ is a zero-mean wide sense stationary (WSS) random process. In this case the sampled signal $x(n)$ is also zero-mean WSS. Let $S_x(e^{j\omega})$ denote its power spectral density. It is shown in Appendix B that $J(\hat{\delta})$ admits the approximation

$$\begin{aligned} J(\hat{\delta}) &\approx \frac{1}{2} \{ [(\delta_1 + \delta_3 - \delta_2) - (\hat{\delta}_1 + \hat{\delta}_3 - \hat{\delta}_2)]^2 A \\ &+ [((\delta_1 - \delta_3) - (\hat{\delta}_1 - \hat{\delta}_3))^2 + (\delta_2 - \hat{\delta}_2)^2] B \\ &+ |\mathbf{c}(\hat{\delta})|^2 C \} \end{aligned} \quad (3.47)$$

where

$$A = \frac{1}{2\pi} \int_{-\alpha\pi}^{\alpha\pi} \omega^2 S_x(e^{j\omega}) d\omega \quad (3.48)$$

$$\frac{B}{2} = \frac{1}{2\pi} \int_{\frac{\pi}{2}-\alpha\pi}^{\frac{\pi}{2}+\alpha\pi} \omega^2 S_x(e^{j\omega}) d\omega \quad (3.49)$$

$$C = 2\alpha N_0. \quad (3.50)$$

In (3.47) all cubic products of entries of δ and $\hat{\delta}$ have been neglected.

The expression (3.47) shows that when the steering vector $\mathbf{c}(\delta)$ satisfies the normalization constraint (3.8) exactly, the function $J(\hat{\delta})$ is minimized by $\hat{\delta} = \delta$ as long as both A and B are strictly positive, which means that $x(n)$ must contain some power in the frequency bands $[-\alpha\pi, \alpha\pi]$ and $[\pm\pi/2 - \alpha\pi, \pm\pi/2 + \alpha\pi]$. Note that conditions that require that the input signal should be rich enough are actually quite common in the system identification literature [31]. When we employ the first-order approximation (3.33) for the steering vector $\mathbf{c}(\delta)$, the squared Euclidean vector norm $|\mathbf{c}(\hat{\delta})|^2$ contains quadratic terms of the timing offset estimates. In this case, small bias terms are introduced in the estimates, which can be neglected as long as

$$A/C \gg 1; \quad B/C \gg 1. \quad (3.51)$$

So, in practice, to ensure a satisfactory performance of the blind calibration method described above, conditions (3.51) need to be satisfied. These conditions are quite interesting, since [32, ch. 3] C/A and C/B represent respectively the Cramer-Rao lower bounds for estimating a timing delay based on the frequency content of $x(n)$ over the bands $[-\alpha\pi, \alpha\pi]$ and $[\pm\pi/2 - \alpha\pi, \pm\pi/2 + \alpha\pi]$, when the signal is observed in additive WGN with variance N_0 .

At this point it is worth observing that the blind calibration scheme does not really require that the signal should be WSS and satisfy conditions (3.51) for all times. As was observed earlier, expression (3.20) for the Fourier transform of the signal

component $r(m, \hat{\delta})$ ensures that locally the error signal energy or power is minimized when the time delay vector is estimated correctly. So the conclusions of the above analysis remain valid even if the input signal is nonstationary, provided the signal $x(n)$ is rich enough to ensure that the quadratic surface approximating the error noise power $J(\hat{\delta})$ in the vicinity of $\hat{\delta}$ has a positive curvature in all directions. Simulations indicate that A/C and B/C do not need to be large for all time segments of the input signal $x(n)$. What is required is that these conditions should hold intermittently over data blocks sufficiently long to allow the quadratic coefficients of A and B in (3.47) to be driven to zero. In other words, if A/C is much greater than 1, but not B/C , for a certain time window, then the quadratic error coefficient of A is driven to zero over this time window. Afterwards it will remain close to zero for a while, so if $B/C \gg 1$ over another time segment, its error term will be minimized, yielding a correct estimate of the delay. However, sporadic excitation is required over the two frequency bands centered at 0 and $\pm\pi/2$, since otherwise, the estimates will slowly drift away from the correct values under the influence of the observation noise. So in conclusion, the conditions (3.51) need to hold only intermittently to ensure that the timing offset values are acquired and periodically refreshed.

IV. CONVERGENCE ANALYSIS

To analyze the adaptive calibration algorithm specified by (3.45) and (3.43) we employ the stochastic averaging or ODE method described in [33] and then developed in full generality in [25]. Let

$$\mathbf{a}(m) = \begin{bmatrix} a_1(m) \\ a_2(m) \\ a_3(m) \\ a_4(m) \end{bmatrix}; \quad \mathbf{b}(m) = \begin{bmatrix} b_2(m) \\ b_3(m) \\ b_4(m) \end{bmatrix}. \quad (4.1)$$

We assume that the step size μ is small, and that the signal $x_c(t)$ is zero-mean WSS. In this case, the vector process

$$\mathbf{y}(m) = [\mathbf{a}^T(m) \quad \mathbf{b}^T(m)]^T \quad (4.2)$$

is also WSS. The averaging method requires the additional assumption that $\mathbf{y}(m)$ can be expressed as

$$\mathbf{y}(m) = f(\xi(m)) \quad (4.3)$$

where $\xi(m)$ is a Markov process. This assumption is satisfied for most practical situations of interest, in particular when $x_c(t)$ is a WSS signal formed by several sinusoidal components, or when it is bandlimited WGN.

Following [25, ch. 2], we write the adaptive algorithm (3.45) as

$$\hat{\delta}(m+1) = \hat{\delta}(m) + \mu \mathbf{T}(\hat{\delta}(m), \mathbf{y}(m)) \quad (4.4)$$

with

$$\mathbf{T}(\hat{\delta}, \mathbf{y}(m)) = \left(\begin{bmatrix} b_2(m) \\ -b_3(m) \\ b_4(m) \end{bmatrix} - \frac{\pi}{4} \begin{bmatrix} -a_2(m) - a_4(m) + 2a_3(m) \\ a_2(m) - a_4(m) \\ a_2(m) + a_4(m) - 2a_3(m) \end{bmatrix} \right) e(m, \hat{\delta}). \quad (4.5)$$

We need first to evaluate

$$t(\hat{\delta}) = E[\mathbf{T}(\hat{\delta}, \mathbf{y}(m))]. \quad (4.6)$$

Since

$$T(\hat{\delta}, \mathbf{y}(m)) = -\nabla_{\hat{\delta}} e^2(m, \hat{\delta})/2 \quad (4.7)$$

we have

$$t(\hat{\delta}) = -\nabla_{\hat{\delta}} J(\hat{\delta}) \quad (4.8)$$

where $J(\hat{\delta})$ is given by (3.39). Consequently $J(\hat{\delta})$ is a Lyapunov function for the ODE

$$\frac{d\hat{\delta}}{dt} = t(\hat{\delta})(t) \quad (4.9)$$

since along trajectories of (4.9) we have

$$\frac{dJ}{dt}(\hat{\delta}(t)) = -|\nabla_{\hat{\delta}} J(\hat{\delta}(t))|^2 \quad (4.10)$$

so the trajectories $\hat{\delta}(t)$ of the ODE (4.9) converge to a minimum of $J(\hat{\delta})$. From the quadratic approximation (3.47) for $J(\hat{\delta})$ we conclude that in the region of small δ and $\hat{\delta}$ for which the approximation is valid, the minimum δ_* is unique and is approximately equal to the true vector δ , up to quadratic error terms.

The stochastic averaging or ODE method relates the behavior of the adaptive algorithm (4.4) to that of the ODE (4.9). Let

$$\mathbf{W} = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 0 & -1 \end{bmatrix} \quad (4.11)$$

and let $O(x)$ denote a term proportional to a variable x . By observing that the matrix

$$\nabla_{\delta} t(\delta) = -\mathbf{W} \text{diag}\{A, B, B\} \mathbf{W}^T + O(N_0) + O(\delta) \quad (4.12)$$

is negative definite and thus stable as long as the $O(N_0)$ and $O(\delta)$ correction terms are dominated by the leading term, we find from Theorem 2 of Chapter 3 of [25] that as μ tends to zero and $m \rightarrow \infty$, the parameter vector

$$\hat{\delta}(m) \sim N(\delta_*, \mu \mathbf{P}) \quad (4.13)$$

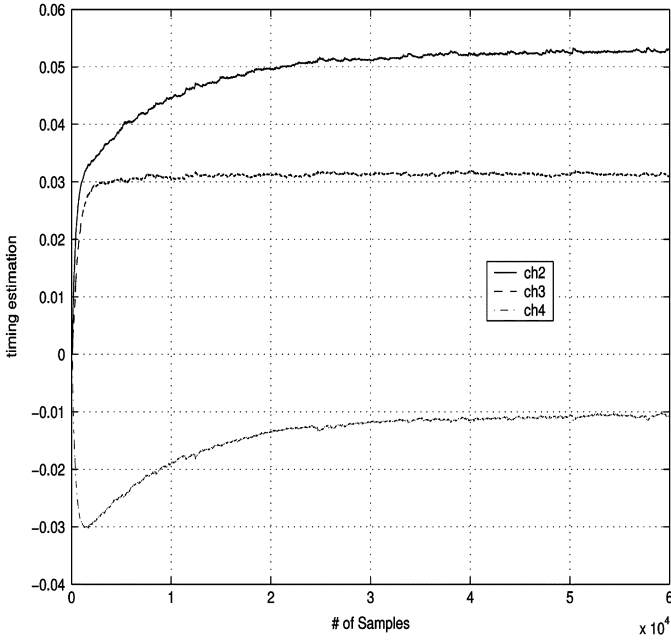


Fig. 7. Convergence of timing offsets for an ADC with $\delta_1 = 0.05$, $\delta_2 = 0.03$, $\delta_3 = -0.01$, a bandlimited WGN input over the band $[-0.75\pi, 0.75\pi]$, SNR = 70 dB, and step size $\mu = 2 \cdot 10^{-4}$.

where $\mathbf{P}$ is a positive definite matrix. This means that by selecting the step size μ sufficiently small, we can ensure that as m becomes large the probability that for a fixed $\epsilon > 0$, $|\hat{\delta}(m) - \delta_*|^2$ is less than ϵ can be made arbitrarily close to one.

V. NUMERICAL SIMULATIONS

In the simulations presented in this section, for $1 \leq i \leq 4$, the filters $k_{i0}(m)$ and $k_{i1}(m)$ are FIR filters obtained by applying a Kaiser window with $\beta = 10$ to the IIR filters with ideal frequency response $K_{i0}(e^{j\omega})$ and $K_{i1}(e^{j\omega})$. To illustrate the performance of the blind calibration method, we consider two examples.

A. Bandlimited White Noise Input

We first consider a bandlimited WGN test signal $x(n)$ extending over the frequency band $[-0.75\pi, 0.75\pi]$. For $1 \leq i \leq 4$, the sequences $x_i(m)$ are obtained by applying time shifts $(i-1) + \delta_{i-1}$ to $x(n)$ with $\delta_0 = 0$, $\delta_1 = 0.05$, $\delta_2 = 0.03$ and $\delta_3 = -0.01$, and then downsampling by a factor of 4, as shown in Fig. 2. The noises $v_i(m)$ are independent zero-mean WGNs such that the signal-to-noise ratio (SNR) at the output of the interleaved ADCs is 70 dB. After sampling, the ADC outputs contain only three alias components over the frequency band $[-\pi, \pi]$. However, since the Kaiser window truncation used to generate the filters $k_{i0}(m)$ and $k_{i1}(m)$ introduces a transition band in their frequency response, we use $4\alpha\pi = 0.9\pi$ in their ideal frequency domain specifications and we use filters with 161 taps. For an adaptation step size $\mu = 2 \cdot 10^{-4}$, the estimates of the timing mismatches are shown in Fig. 7. The estimates obtained are in good agreement with the true values, but are slightly amplified, i.e., the estimates $\hat{\delta}_i = a\delta_i$, where the multiplicative gain a is slightly above one. We believe that this amplification effect is due to the quadratic and higher order delay terms that were neglected in approximations (3.33) and

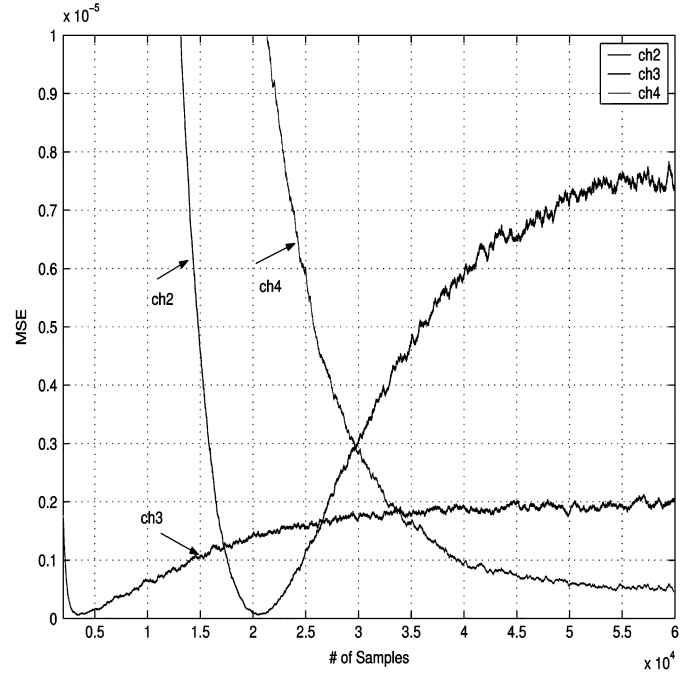


Fig. 8. Sampled mean-square timing offset errors for a Monte Carlo simulation with $K = 100$ trials.

(3.35) for the null-steering vector $\mathbf{c}(\delta)$ and low-pass delay filters $K_i(e^{j\omega}, \hat{\delta}_{i-1})$.

To illustrate the average behavior of the adaptive calibration algorithm over different realizations of the input signal and quantization noises, a Monte Carlo simulation with $K = 100$ trials was performed for the bandlimited WGN signal example considered above. The signal bandwidth, timing offsets, SNR, and adaptation step size μ are the same as before, and for each trial a block of $6 \cdot 10^4$ signal samples is processed. If $\hat{\delta}_i^k(m)$ denotes the estimate of timing offset δ_i obtained at time m for the k th trial, the sampled mean-square error

$$M_i(m) = \frac{1}{K} \sum_{k=1}^K (\hat{\delta}_i^k(m) - \delta_i)^2 \quad (5.1)$$

is plotted as a function of m for $1 \leq i \leq 3$ in Fig. 8. Since the errors at the start of the algorithm are large, we only show the mean-square error values between $m = 2000$ and $6 \cdot 10^4$. In this respect, it is worth pointing out that as indicated in (4.13), the asymptotic MSE contains two components: a squared bias term $(\delta_{*i} - \delta_i)^2$ and an excess mean-square error term proportional to μ . This second term, which is common to all adaptive algorithms, can be decreased by lowering the adaptation step size μ . Of course, lowering μ increases the convergence time of the adaptive algorithm, so a reasonable strategy would probably consist of switching to a lower value of μ after the algorithm has converged.

B. Sinusoidal Input Signal

For the second test example, we use a test signal formed by three sinusoidal components with unit amplitude at frequencies $\omega_1 = 0.1\pi$, $\omega_2 = 0.45\pi$ and $\omega_3 = 0.75\pi$. The timing offsets δ_i are the same as in the first example and the SNR equals 70 dB.

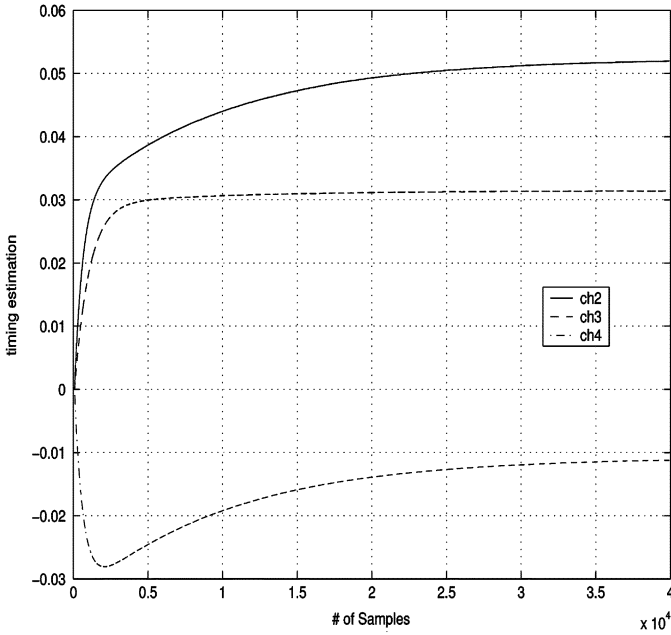


Fig. 9. Convergence of timing offsets for an ADC with $\delta_1 = 0.05$, $\delta_2 = 0.03$, $\delta_3 = -0.01$, an input with three sinusoidal components at $\omega_1 = 0.1\pi$, $\omega_2 = 0.45\pi$, and $\omega_3 = 0.75\pi$, SNR = 70 dB, and step size $\mu = 10^{-3}$.

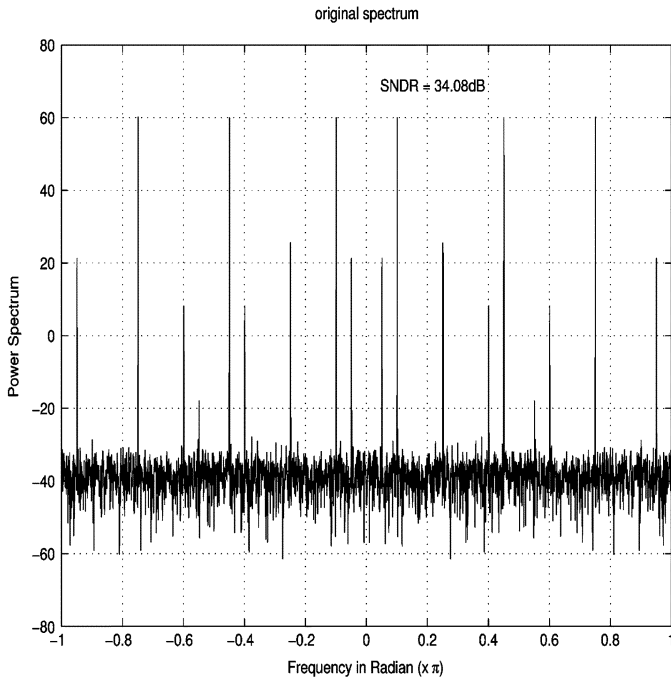


Fig. 10. PSD of the uncalibrated ADC output with 70 dB SNR, $\delta_1 = 0.05$, $\delta_2 = 0.03$, $\delta_3 = -0.01$, for an input with three sinusoidal components at $\omega_1 = 0.1\pi$, $\omega_2 = 0.45\pi$, and $\omega_3 = 0.75\pi$.

For this case we select $4\alpha\pi = 0.6\pi$ in the frequency domain specification of the filters $K_{i0}(e^{j\omega})$ and $K_{i1}(e^{j\omega})$, and the FIR filters have 81 taps. For an adaptation step size $\mu = 10^{-3}$, the estimates of the timing mismatches are shown in Fig. 9. Again, a small amplification factor can be observed in the obtained estimates $\hat{\delta}_i$.

To illustrate the performance of the Farrow correction filters specified by (2.17), for the three sinusoid example considered above, we compare the performance of the uncorrected ADC corresponding to the case where $G_i(z) = z^{-(i-1)}$ in Fig. 3

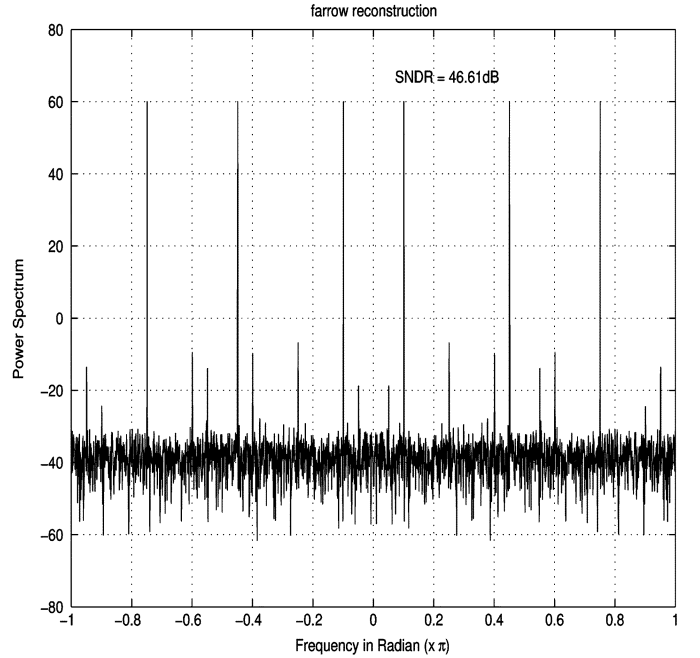


Fig. 11. PSD of the corrected ADC output with 70 dB SNR, $\delta_1 = 0.05$, $\delta_2 = 0.03$, $\delta_3 = -0.01$, for an input with three sinusoidal components at $\omega_1 = 0.1\pi$, $\omega_2 = 0.45\pi$, and $\omega_3 = 0.75\pi$.

to that of a corrected ADC. The PSD of the output of the uncorrected ADC is shown in Fig. 10. The signal to noise and distortion ratio (SNDR) of the uncorrected sampled signal is 34.08 dB.

In contrast, Fig. 11 shows the PSD of the reconstructed signal $\hat{x}(n)$ obtained by using the Farrow correction filters with the steady-state estimated timing offsets of Fig. 9. Because of the presence of discontinuities at $\omega = \pm\pi/2$ we employ FIR filters with 161 taps to implement the correction filters $S_e(e^{j\omega})$, $jS_o(e^{j\omega})$, and $j\Psi(e^{j\omega})$. These filters are designed by first evaluating their ideal infinite impulse responses in closed-form and then by applying a Kaiser window with $\beta = 10$. Note however that since the filters operate on sequences that have been up-sampled by a factor of 4, only 1/4th of the coefficients are used at each time instant. In effect, the correction filters are periodic with length 41 and period $M = 4$. As indicated by Fig. 11, a small amount of aliasing is still present in the corrected output, but the signal-to-noise-distortion ratio (SNDR) is now 46.61 dB, so a SNDR improvement of over 12 dB has been achieved by calibration.

It is worth noting that alias components located close to the filter discontinuities at $\pm\pi/2$, such as those at 0.4π and 0.6π , are attenuated less by the correction filters than alias components located further away from filter discontinuities.

VI. CONCLUSION

A new blind calibration method has been proposed for estimating the timing offsets of four-channel time-interleaved ADCs. This method requires that the input signal should be slightly oversampled. This creates a low frequency band where the outputs of the interleaved ADCs contain only three alias components, instead of four. The matrix PSD of the ADC sampling channels is therefore rank deficient over this frequency band. This property is exploited by constructing a null steering

filter bank which is parametrized in terms of the unknown timing offsets. An adaptive stochastic gradient method is used to estimate the timing offsets, whose convergence is analyzed. Whereas the blind calibration method described here for $M = 4$ channels differs significantly from the method developed in [20] for two channels, which took the form of a time delay estimation problem, as shown in [34] there is no conceptual difficulty in extending the null steering scheme described here to an arbitrary number M of channels. However, when M is arbitrary, it is no longer possible to derive closed form expressions of the form (2.17) for the correction filters $G_i(e^{j\omega})$. So the correction filters need to be evaluated numerically by employing the approach described in [13] to reconstruct a uniformly sampled signal from periodically sampled nonuniform sequences (see also [14]). Also, for M arbitrary, it is no longer possible to derive a closed-form expression for the null-steering vector $\mathbf{c}^T(\boldsymbol{\delta})$ solving (3.6). Instead, symbolic computation software (such as Mathematica) needs to be used to derive a first-order expansion of the form (3.34) for the null-steering vector.

APPENDIX A

FARROW CORRECTION OF TIMING OFFSETS

To construct an approximate inverse of the matrix $M(e^{j\omega}, \boldsymbol{\delta})$ given by (2.12)–(2.14), observe that for small $\boldsymbol{\delta}$, the Vandermonde matrix $\mathbf{V}(\boldsymbol{\delta})$ admits the linear approximation

$$\mathbf{V}(\boldsymbol{\delta}) = 2\boldsymbol{\Omega} + \mathbf{L}(\boldsymbol{\delta}) + O(|\boldsymbol{\delta}|^2) \quad (\text{A.1})$$

where $\boldsymbol{\Omega} = V(\mathbf{0})/2$ is the discrete Fourier transform matrix of size 4, i.e.,

$$\boldsymbol{\Omega} = (\omega_{ik}, 1 \leq i, k \leq 4); \quad \omega_{ik} = \frac{1}{2}(\omega_4)^{(i-1)(k-1)} \quad (\text{A.2})$$

with $\omega_4 = e^{-j\pi/2}$, and

$$\mathbf{L}(\boldsymbol{\delta}) = \frac{\pi}{2} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -\delta_1 & 2j\delta_1 & 3\delta_1 \\ 0 & j\delta_2 & -2j\delta_2 & 3j\delta_2 \\ 0 & \delta_3 & 2j\delta_3 & -3\delta_3 \end{bmatrix}. \quad (\text{A.3})$$

Note that $\boldsymbol{\Omega}$ is symmetric and unitary, i.e., $\boldsymbol{\Omega}^{-1} = \boldsymbol{\Omega}^H = \boldsymbol{\Omega}^*$, where $\boldsymbol{\Omega}^*$ denotes the complex conjugate of $\boldsymbol{\Omega}$. Then by inverting (2.9) we find

$$\mathbf{X}_A(e^{j\omega}) = 4\mathbf{V}^{-1}(\boldsymbol{\delta})\mathbf{D}^{-1}(e^{j\omega}, \boldsymbol{\delta})\mathbf{X}_I(e^{j\omega}) \quad (\text{A.4})$$

where

$$\mathbf{D}^{-1}(e^{j\omega}, \boldsymbol{\delta}) = \left(\mathbf{I}_4 - \frac{j\omega}{4} \text{diag}\{0, \delta_1, \delta_2, \delta_3\} \right) \mathbf{D}_0(e^{j\omega}) + O(\boldsymbol{\delta}^2) \quad (\text{A.5})$$

with

$$\mathbf{D}_0(e^{j\omega}) = \text{diag}\left\{e^{j(i-1)\omega/4}, 1 \leq i \leq 4\right\} \quad (\text{A.6})$$

and

$$4\mathbf{V}^{-1}(\boldsymbol{\delta}) = 2\boldsymbol{\Omega}^H - \boldsymbol{\Gamma}(\boldsymbol{\delta}) + O(\boldsymbol{\delta}^2) \quad (\text{A.7})$$

with

$$\boldsymbol{\Gamma}(\boldsymbol{\delta}) = \boldsymbol{\Omega}^H \mathbf{L}(\boldsymbol{\delta}) \boldsymbol{\Omega}^H = \sum_{i=1}^3 \boldsymbol{\Gamma}_i \delta_i \quad (\text{A.8})$$

where

$$\begin{aligned} \boldsymbol{\Gamma}_1 &= \frac{\pi}{4} \begin{bmatrix} 1+j & -3j & -1+j & j \\ -1+j & 3 & -(1+j) & -1 \\ -(1+j) & 3j & 1-j & -j \\ 1-j & -3 & 1+j & 1 \end{bmatrix} \\ \boldsymbol{\Gamma}_2 &= \frac{\pi}{4} \begin{bmatrix} j & 1+j & -3j & -1+j \\ -j & -(1+j) & 3j & 1-j \\ j & 1+j & -3j & -1+j \\ -j & -(1+j) & 3j & 1-j \end{bmatrix} \\ \boldsymbol{\Gamma}_3 &= \frac{\pi}{4} \begin{bmatrix} -1+j & j & 1+j & -3j \\ 1+j & 1 & 1-j & -3 \\ 1-j & -j & -(1+j) & 3j \\ -(1+j) & -1 & -1+j & 3 \end{bmatrix}. \end{aligned} \quad (\text{A.9})$$

Substituting (A.7) and (A.9) inside (A.4) yields

$$\begin{aligned} \mathbf{X}_A(e^{j\omega}) &= [2\boldsymbol{\Omega}^H - \frac{j\omega}{2}\boldsymbol{\Omega}^H \text{diag}\{0, \delta_1, \delta_2, \delta_3\} \\ &\quad - \boldsymbol{\Gamma}(\boldsymbol{\delta})] \mathbf{D}_0^{-1}(e^{j\omega}) \mathbf{X}_I(e^{j\omega}). \end{aligned} \quad (\text{A.10})$$

Next is we replace $\omega/4$ (resp. $\omega/4 - \pi/2$, $\omega/4 - \pi$, and $\omega/4 - 3\pi/2$) by ω with $\pi/2 \leq \omega < \pi$ (resp. $0 \leq \omega < \pi/2$, $-\pi/2 \leq \omega < 0$, and $-\pi \leq \omega < -\pi/2$) in the first row (resp. second, third, and fourth row) of (A.10), we obtain the reconstruction expression (2.16) where the filters $G_i(e^{j\omega})$ are given by (2.17) and evaluated over the range $\pi/2 \leq \omega < \pi$ (resp. over $0 \leq \omega < \pi/2$, $-\pi/2 \leq \omega < 0$, and $-\pi < \omega < -\pi/2$).

APPENDIX B

QUADRATIC APPROXIMATION OF $J(\hat{\boldsymbol{\delta}})$

When the fast sampled sequence $x(n)$ is zero-mean WSS with autocorrelation $R_x(n)$ and PSD $S_x(e^{j\omega})$, by using the model (2.4), we find that the interleaved sequences $x_i(m)$ are jointly WSS with auto/cross-correlation

$$\begin{aligned} R_{ik}(\ell - m) &= E[x_i(\ell)x_k(m)] \\ &= (f_i * R_x * f_k^R)(4(\ell - m)) \end{aligned} \quad (\text{B.1})$$

where $f_i^R(n) = f_i(-n)$. The cross-spectral density of $x_i(m)$ and $x_k(m)$ is given by

$$\begin{aligned} S_{ik}(e^{j\omega}) &= \frac{1}{4} \sum_{p=1}^4 H_i(e^{j(\omega-(p-1)2\pi)/4}) \\ &\quad \times S_x(e^{j(\omega-(p-1)2\pi)/4}) H_k^*(e^{j(\omega-(p-1)2\pi)/4}), \end{aligned} \quad (\text{B.2})$$

so that the PSD of the vector process

$$\mathbf{x}_I(m) = [x_1(m) \quad x_2(m) \quad x_3(m) \quad x_4(m)]^T \quad (\text{B.3})$$

can be expressed as

$$\mathbf{S}_{\mathbf{x}_I}(e^{j\omega}) = \frac{1}{4} \mathbf{M}(e^{j\omega}, \boldsymbol{\delta}) \mathbf{S}_A(e^{j\omega}) \mathbf{M}^H(e^{j\omega}, \boldsymbol{\delta}) \quad (\text{B.4})$$

where $\mathbf{M}(e^{j\omega}, \delta)$ is given by (2.10) and

$$\mathbf{S}_A(e^{j\omega}) = \text{diag}\left\{S_x\left(e^{j\frac{\omega}{4}}\right), S_x\left(e^{j\left(\frac{\omega}{4}-\frac{\pi}{2}\right)}\right), S_x\left(e^{j\left(\frac{\omega}{4}-\pi\right)}\right), S_x\left(e^{j\left(\frac{\omega}{4}-3\frac{\pi}{2}\right)}\right)\right\}. \quad (\text{B.5})$$

Furthermore, over the frequency band $[-4\alpha\pi, 4\alpha\pi]$, one of the alias components is missing and (B.4) becomes

$$\mathbf{S}_{\mathbf{x}_I}(e^{j\omega}) = \frac{1}{4}\mathbf{M}_R(e^{j\omega}, \delta)\mathbf{S}_R(e^{j\omega})\mathbf{M}_R^H(e^{j\omega}, \delta) \quad (\text{B.6})$$

where $\mathbf{M}_R(e^{j\omega}, \delta)$ is given by (3.4) and

$$\mathbf{S}_R(e^{j\omega}) \triangleq \text{diag}\left\{S_x\left(e^{j\left(\frac{\omega}{4}+\frac{\pi}{2}\right)}\right), S_x\left(e^{j\frac{\omega}{4}}\right), S_x\left(e^{j\left(\frac{\omega}{4}-\frac{\pi}{2}\right)}\right)\right\}. \quad (\text{B.7})$$

Then, if we consider the decomposition (3.18) of the error signal $e(m, \hat{\delta})$ in its signal and noise components, the PSD of the signal component $r(m, \hat{\delta})$ is given by

$$\begin{aligned} S_r(e^{j\omega}) &= \mathbf{H}(e^{j\omega}, \hat{\delta})\mathbf{S}_{\mathbf{x}_I}(e^{j\omega})\mathbf{H}^H(e^{j\omega}, \hat{\delta}) \\ &= \frac{1}{4}\mathbf{c}^T(\hat{\delta})\mathbf{E}(e^{j\omega}, \delta - \hat{\delta})\mathbf{V}_R(\delta)\mathbf{S}_R(e^{j\omega}) \\ &\quad \times \mathbf{V}_R^H(\delta)\mathbf{E}^H(e^{j\omega}, \delta - \hat{\delta})\mathbf{c}(\hat{\delta}) \end{aligned} \quad (\text{B.8})$$

for $0 \leq |\omega| < 4\alpha\pi$ and $S_r(e^{j\omega}) = 0$ otherwise. To evaluate (B.8), we perform the first-order Taylor series approximations

$$\mathbf{E}(e^{j\omega}, \delta - \hat{\delta}) \approx \mathbf{I}_4 + \frac{j\omega}{4}\text{diag}\{\epsilon_i, 1 \leq i \leq 4\}. \quad (\text{B.9})$$

and

$$\mathbf{c}(\hat{\delta}) \approx \mathbf{c}(\delta) + \frac{\pi}{4} \begin{bmatrix} 0 \\ \epsilon_1 - \epsilon_2 - \epsilon_3 \\ 2(\epsilon_3 - \epsilon_1) \\ \epsilon_1 + \epsilon_2 - \epsilon_3 \end{bmatrix} \quad (\text{B.10})$$

of the matrix $\mathbf{E}(e^{j\omega}, \delta - \hat{\delta})$ given by (3.21) and of the vector $\mathbf{c}(\hat{\delta})$, where in (B.9)–(B.10), we denote by

$$\epsilon_i = \delta_i - \hat{\delta}_i \quad (\text{B.11})$$

the deviation between the timing offset δ_i and its estimate $\hat{\delta}_i$. Then by noting that

$$\mathbf{c}^T(\delta) = [1 \quad -1 \quad 1 \quad -1] + O(\delta) \quad (\text{B.12})$$

and

$$\mathbf{V}_R(\delta) = \mathbf{\Omega}_R + O(\delta) \quad (\text{B.13})$$

with

$$\mathbf{\Omega}_R \triangleq \begin{bmatrix} 1 & 1 & 1 \\ j & 1 & -j \\ -1 & 1 & -1 \\ -j & 1 & j \end{bmatrix} \quad (\text{B.14})$$

we find

$$\begin{aligned} &\mathbf{V}_R^H(\delta)\mathbf{E}^H(e^{j\omega}, \delta - \hat{\delta})\mathbf{c}(\hat{\delta}) \\ &\approx \mathbf{\Omega}_R^H \begin{bmatrix} 0 \\ j\frac{\omega}{4}\epsilon_1 + \frac{\pi}{4}(\epsilon_1 - \epsilon_2 - \epsilon_3) \\ -j\frac{\omega}{4}\epsilon_2 + \frac{\pi}{2}(-\epsilon_1 + \epsilon_3) \\ j\frac{\omega}{4}\epsilon_3 + \frac{\pi}{4}(\epsilon_1 + \epsilon_2 - \epsilon_3) \end{bmatrix} \\ &= \begin{bmatrix} \left(\frac{\omega}{4} + \frac{\pi}{2}\right)(\epsilon_1 - \epsilon_3 + j\epsilon_2) \\ j\frac{\omega}{4}(\epsilon_1 + \epsilon_3 - \epsilon_2) \\ \left(\frac{\omega}{4} - \frac{\pi}{2}\right)(-\epsilon_1 + \epsilon_3 + j\epsilon_2) \end{bmatrix} \end{aligned} \quad (\text{B.15})$$

where in (B.15) the error terms are quadratic in δ and $\hat{\delta}$. Substituting inside (B.8) gives

$$\begin{aligned} S_r(e^{j\omega}) &\approx \frac{1}{4} \left\{ (\epsilon_1 + \epsilon_3 - \epsilon_2)^2 \left(\frac{\omega}{4}\right)^2 S_x\left(e^{j\frac{\omega}{4}}\right) \right. \\ &\quad + [(\epsilon_1 - \epsilon_3)^2 + \epsilon_2^2] \left[\left(\frac{\omega}{4} + \frac{\pi}{2}\right)^2 S_x\left(e^{j\left(\frac{\omega}{4}+\frac{\pi}{2}\right)}\right) \right. \\ &\quad \left. \left. + \left(\frac{\omega}{4} - \frac{\pi}{2}\right)^2 S_x\left(e^{j\left(\frac{\omega}{4}-\frac{\pi}{2}\right)}\right)\right] \right\} \end{aligned} \quad (\text{B.16})$$

for $0 \leq |\omega| < 4\alpha\pi$ and $S_r(e^{j\omega}) = 0$ otherwise, where all cubic products of the entries of δ and $\hat{\delta}$ have been neglected.

The PSD of the noise component $w(m, \hat{\delta})$ of $e(m, \hat{\delta})$ is given by (3.24). Substituting (B.16) and (3.24) inside

$$\begin{aligned} J(\hat{\delta}) &= \frac{1}{2}E[e^2(m, \hat{\delta})] \\ &= \frac{1}{4\pi} \int_{-\pi}^{\pi} [S_r(e^{j\omega}) + S_w(e^{j\omega})] d\omega \end{aligned} \quad (\text{B.17})$$

gives expressions (3.47)–(3.50).

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